**Your Name:**

**ID #:**

## Worksheet: Limits to Infinity-Part 1

**Note: for all the problems, identify the limit as a number, $-\infty$, $\infty$ or DNE (does not exist)**

1. Please keep in mind the list of functions in order of their rate of growth – quickest to slowest:

   **Factorial**( x! ), **Exponential** ( $4^x$, $e^x$ ), **Algebraic** ($x^4$, $x^{0.5}$), **Logarithm** ( $log_2 x$, $lnx$ )

   Then find the following limits:

   (a) $\lim\limits_{x\to\infty} \dfrac{x^3}{x!} =$

   (b) $\lim\limits_{x\to\infty} \dfrac{\sqrt[3]{x^4+2x^3+1}}{lnx+3} =$

   (c) $\lim\limits_{x\to\infty} \dfrac{x^{10}+4x^7+10}{32x^{12}+10} =$

   (d) $\lim\limits_{x\to\infty} \dfrac{5x+1}{x^2-3x+4}$

   (e) $\lim\limits_{x\to-\infty} \dfrac{2^x}{x^2}$

   (f) $\lim\limits_{x\to\infty} \dfrac{2^x}{x^2}$

   (g) $\lim\limits_{n\to\infty} \dfrac{5^n+2}{n^9+100} =$

   (h) $\lim\limits_{n\to\infty} \dfrac{n^4+10lnn}{4\sqrt{n^8+3n+100}} =$

   (i) $\lim\limits_{n\to\infty} \dfrac{-n^4+3n^2}{19n^3+3n^2-10} =$

   (j) $\lim\limits_{x\to-\infty} \dfrac{x^{120}+20}{-x^{100}+e^x} =$

   (k) $\lim\limits_{n\to-\infty} \dfrac{5^n+n^2}{-e^n+1} =$

   (l) $\lim\limits_{x\to\infty} \dfrac{x^8+lnx}{3x^8+2} =$

   (m) $\lim\limits_{x\to\infty} \dfrac{x^{10}+2x^2+10}{32x^6+12} =$

   (n) $\lim\limits_{n\to\infty} \dfrac{n^5-10n^7+lnn}{n^3+8n^7} =$

   (o) $\lim\limits_{x\to\infty} \dfrac{e^x}{x^3}$

   (p) $\lim\limits_{x\to-\infty} (-3x^3-4x+5)$

   (q) $\lim\limits_{n\to\infty} \dfrac{n^{15}-10n^7+lnn}{n^3+8n^7} =$

2. Show all the work to find limits for some common functions:

(a) $\lim_{x \to -\infty} e^x =$

(b) $\lim_{x \to -\infty} xlnx =$

(c) $\lim_{x \to \infty} xlnx =$

(d) $\lim_{x \to -\infty} e^{-x} =$

(e) $\lim_{x \to \infty} e^{-x} =$

(f) $\lim_{x \to \infty} \frac{1}{x} \sin x =$

(g) $\lim_{x \to -\infty} e^{\frac{1}{x}} =$

(h) $\lim_{x \to \infty} e^{\frac{1}{x}} =$

(i) $\lim_{x \to \infty} \frac{1}{e^{-x}} =$

(j) $\lim_{x \to \infty} lnx =$

(k) $\lim_{x \to \infty} \sin x =$

(l) $\lim_{x \to \infty} \sin(\frac{1}{x}) =$